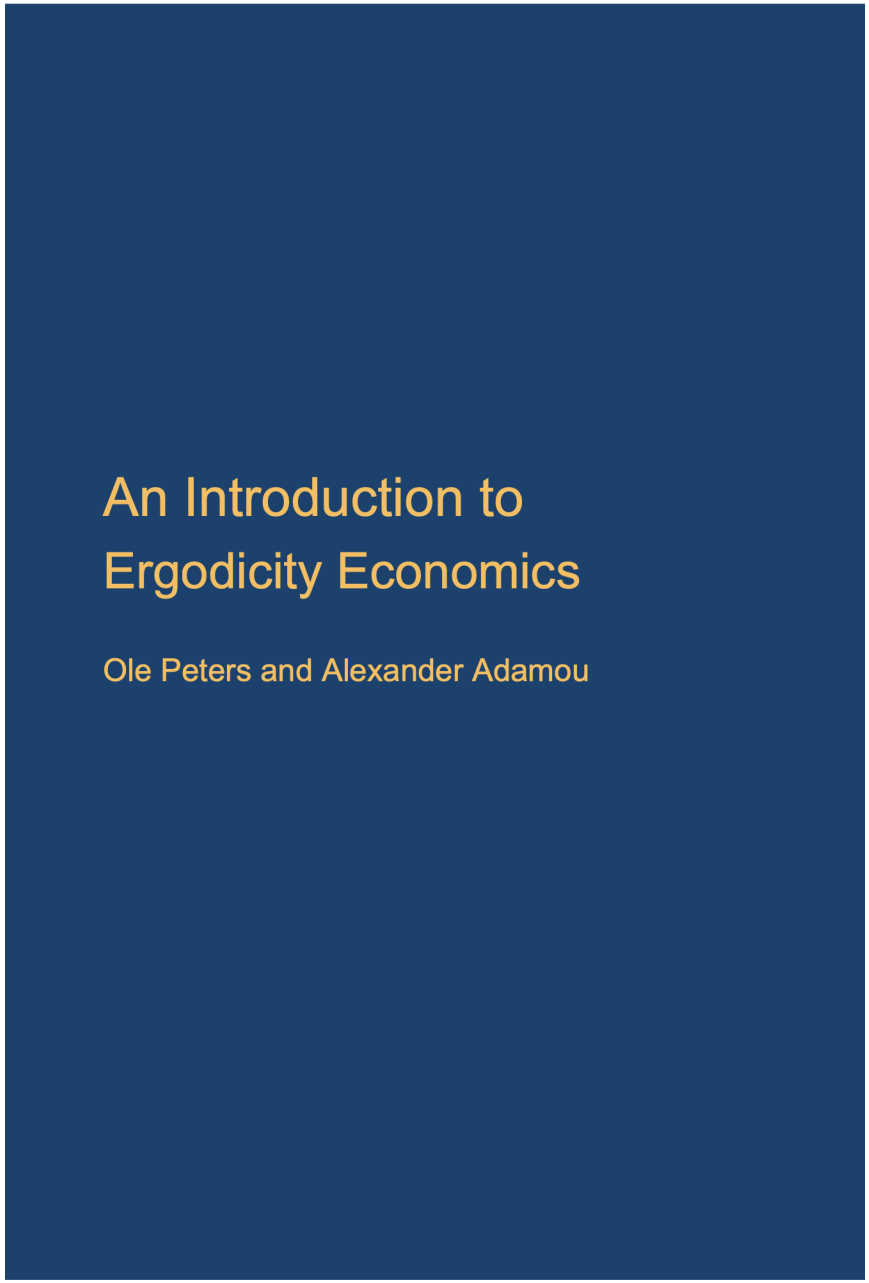
An Introduction to
Ergodicity Economics
Ole Peters and Alexander Adamou

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Scope of the book

Coin tosses

Ergodicities

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Long-term growth

Axiom: Agents maximize wealth.

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Interesting because

- Ergodicity broken in standard models in economics $\rightarrow$ expected wealth $\neq$ real wealth over time.
- Standard analyses in economics use expected wealth.
- Re-do those analyses $\rightarrow$ different worldview.

Coin tosses

- The infamous coin toss
- The leveraged coin toss
- The cooperating coin toss
- The social coin toss

Coin tosses

- **The infamous coin toss**
- The leveraged coin toss
- The cooperating coin toss
- The social coin toss

The infamous coin toss

Let us define our model of the dynamics of your wealth under the rules we just specified.[1] At regular intervals of duration δt, we randomly generate a factor r with each possible value $r_i \in \{0.6, 1.5\}$ occurring with probability $1/2$,

$$r = \begin{cases} 0.6 & \text{with probability } 1/2 \\ 1.5 & \text{with probability } 1/2 \end{cases} \tag{1.1}$$

and multiply current wealth by that factor, so that

$$x(t + \delta t) = x(t)r. \tag{1.2}$$

Chapter 1.1., p.2.

The infamous coin toss

Comments:

- discrete Geometric Brownian Motion (GBM): finance theory.

The infamous coin toss

Comments:

- discrete Geometric Brownian Motion (GBM): finance theory.
- multiplicative: evolution, life, investing, learning, growing.

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Simple ergodicity problem:

$$\text{expected value} \neq \text{time average}.$$

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Philosophy: keep the text light.

Simple ergodicity problem:

$$\text{expected value} \neq \text{time average}.$$

But $\approx$150 years of theory behind this.

Dynamical systems

Applications often use *stochastic* processes.
Ergodic theory written in *deterministic* dynamical systems.
Consistent because DS describe unobservable states.

Dynamical systems

Dynamical system: (X, F, μ, T)

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Mental image:

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$X =$ state space: any thing, x_i, you can be.

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X = state space: any thing, x_i, you can be.

F = field of sets of x_i's. Coarse-graining: you're special, I know little about you. But I may know a set $A \in F$ which you're part of.

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μ measure: tells the size of a set, $\mu(A)$.

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μ measure: tells the size of a set, $\mu(A)$.

T = measure-preserving time evolution operator.
Set A today will be $T(A)$ tomorrow, then $T^2(A)$ *etc.*
Size of a set stays the same, $\mu(T^{-1}A) = \mu(A)$.

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Dynamical systems

- Mathematically, march through time deterministically.
- But not physically visible.
- State x_i deeply hidden, unobservable, "the true you."
- True you unknowable, but we can measure observables,
  $f(x_i)$: how rich are you?
  $f(T(x_i))$: how rich tomorrow?

Ergodic hierarchy

Ergodicity means

- Metric transitivity

Ergodic hierarchy

Ergodicity means

- Metric transitivity
  "any differences are transitory."
  For any sets $A, B \in F$, there exists an n such that $\mu(T^{-n}(A) \cap B) > 0$ (mixing).

  Equivalently, "can't get stuck."
  If A satisfies $T^{-1}(A) = A$, then either $\mu(A) = 1$ or $\mu(A) = 0$ (only trivial invariant sets).

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions
"Measure $f(x_i)$ for a long time; find values with the same frequency as for $f(x_j)$."

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions

$\rightarrow$ Equality of averages

Ergodic hierarchy

Ergodicity means

- Metric transitivity

$\rightarrow$ Equality of distributions

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"Averaging over time is equivalent to averaging over the ergodic ensemble."

Ergodic hierarchy

Ergodicity means

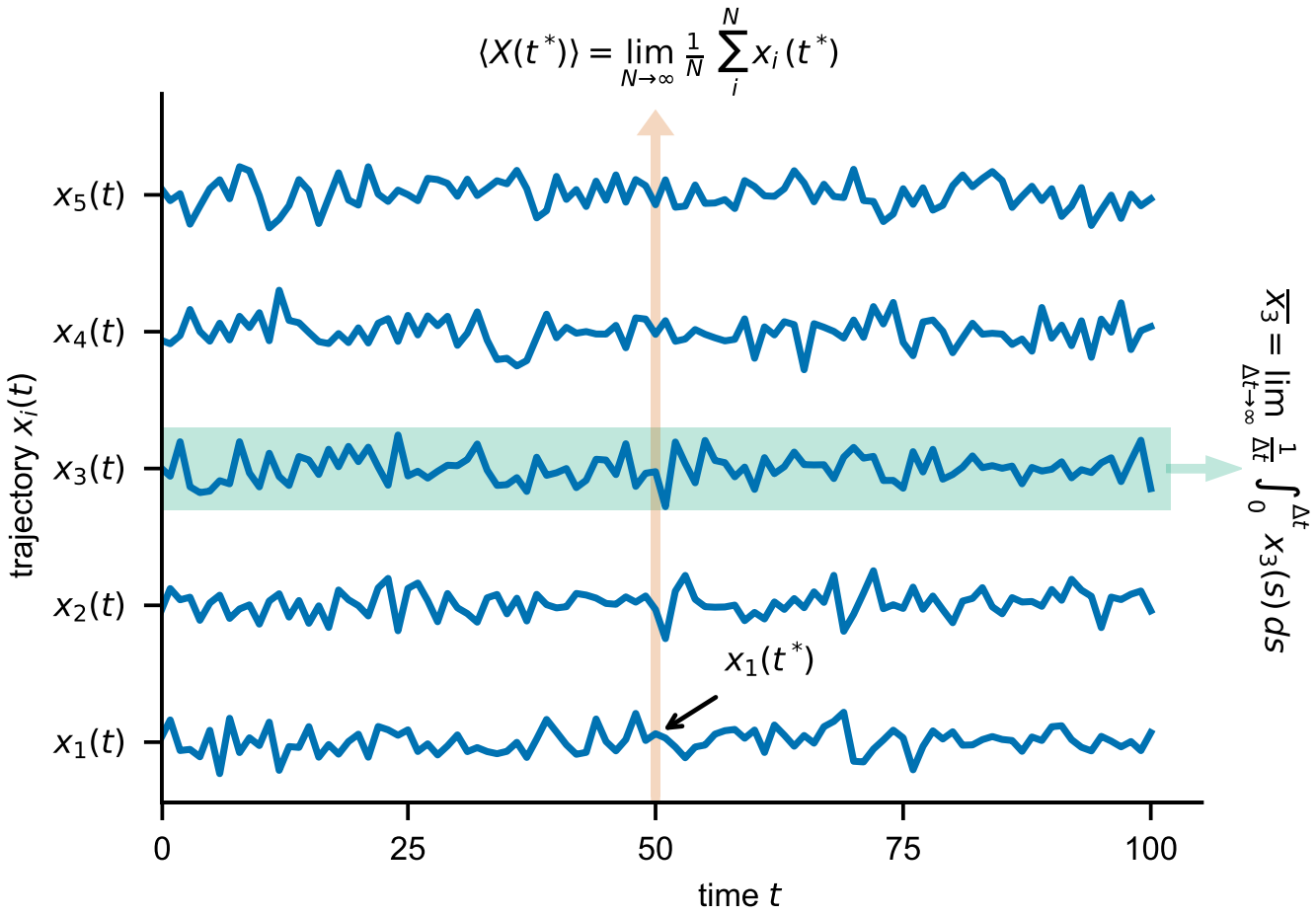
- Metric transitivity

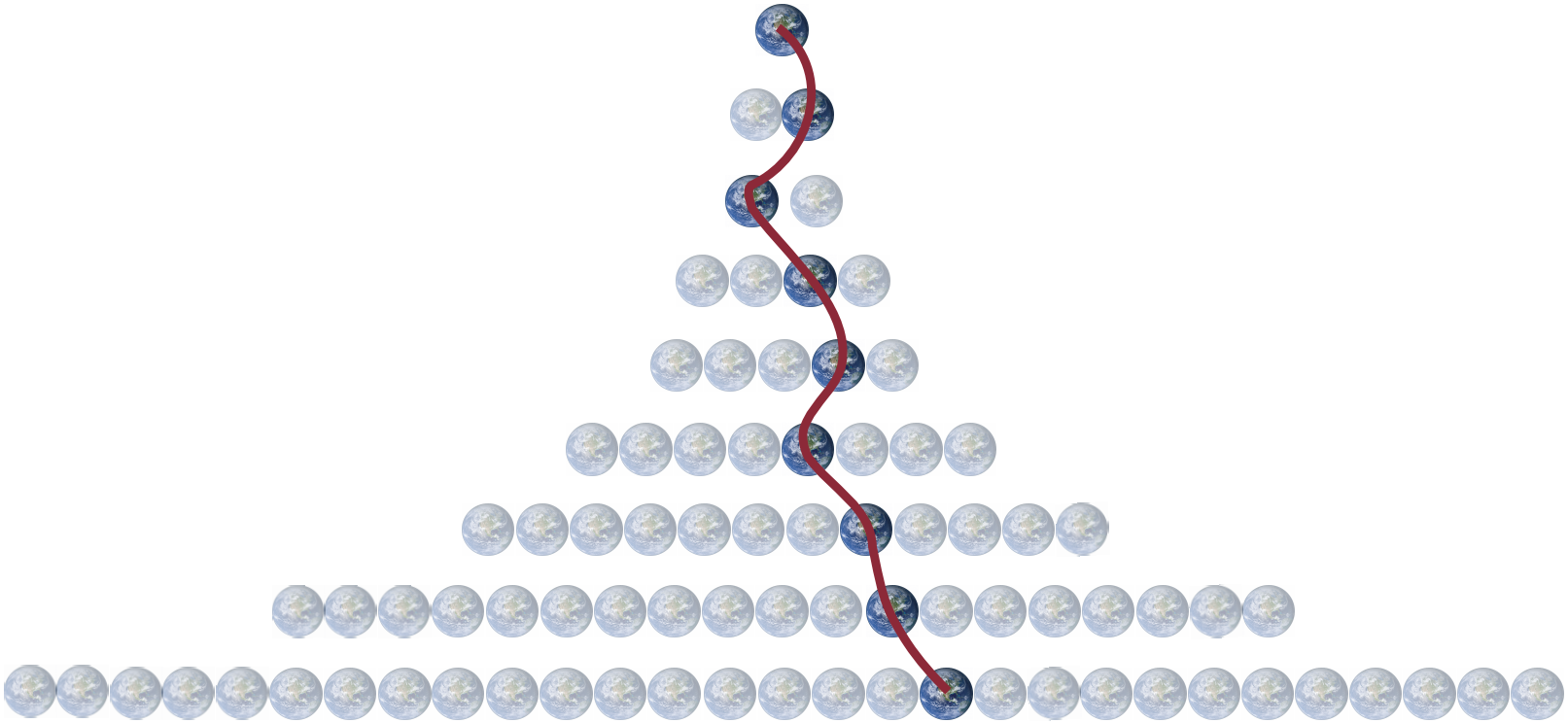
$\rightarrow$ Equality of distributions

$\rightarrow$ Equality of averages

Averages

Averages



Chapter 2, p.9:

Thank you.